

Sphere isometries preserve the Plücker body

MathlibAnnex.PluckerBody.eq_of_sphereIsometry

theorem

Combines generator transport and its inverse for every finite output dimension.

Statement

Let $m, N \in \mathbb{N}_0$, put $n = m + 1 > 0$, and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p, q : E \rightarrow \mathbb{R}$ be continuous real seminorms with specified positive constants a_p, b_p, a_q, b_q such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty \quad (x \in E).$$

Both p and q are norms. For each $s \in \{p, q\}$ define the closed unit ball, unit sphere, and real reference volume by

$$K_s = \{x : s(x) \leq 1\}, \quad \Sigma_s = \{x : s(x) = 1\}, \quad v_s = \lambda_n(K_s) \in (0, \infty).$$

The finite measure is read as a real number when used as a scalar.

Let $\mathcal{I}_{n,N}$ be the finite set of increasing n -tuples of distinct rows from $\{1, \dots, N\}$. The minor-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ carries its sup norm. For a linear map $B : E \rightarrow F$, put $B_{ij} = (Be_j)_i$ in the standard coordinate basis. Write $\Pi_n(B)$ for its maximal-minor vector and $\mathcal{V}_p(B), \mathcal{V}_q(B)$ for its two volume-scaled versions. Their coordinates and scaling are

$$\Pi_n(B)_I = \det(B_{i_a,j})_{a,j=1}^n, \quad \mathcal{V}_s(B) = v_s \Pi_n(B) \quad (s \in \{p, q\}) \quad (I = (i_1 < \dots < i_n)).$$

For each $s \in \{p, q\}$, let $\mathcal{C}_s^N$ consist of linear maps $B : E \rightarrow F$ with $\|Bz\|_\infty \leq s(z)$ for all z . Set

$$\mathcal{G}_s^N = \{\pm \mathcal{V}_s(B) : B \in \mathcal{C}_s^N\}, \quad \mathcal{P}_s^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_s^N).$$

If $\Delta : \Sigma_p \rightarrow \Sigma_q$ is a bijection with $q(\Delta(u) - \Delta(v)) = p(u - v)$ for all $u, v \in \Sigma_p$, then

$$\mathcal{P}_p^N = \mathcal{P}_q^N.$$

Assumptions

The source dimension is positive, $n = m + 1$. Both seminorms have the specified positive norm comparisons and are continuous. The sphere map is a bijective chord-distance isometry. The output dimension N is arbitrary, including zero.

Conclusion

The two bodies agree for each finite N . This statement does not classify the norms or assert that the given sphere map itself has a linear extension.

Proof route

Transport both signs, take convex hulls, and repeat for the inverse isometry.

▼ Proof steps

1. **Place every signed target generator in the source body.** For each $B \in \mathcal{C}_q^N$,

$$\mathcal{V}_q(B) \in \mathcal{P}_p^N, \quad -\mathcal{V}_q(B) \in \mathcal{P}_p^N.$$

The first membership is generator transport through the sphere isometry applied to Δ and this q -contraction B . The second is sign symmetry of the same source body. Consequently,

$$\mathcal{G}_q^N \subseteq \mathcal{P}_p^N.$$

This is the signed-generator inclusion.

2. **Pass to convex hulls.** Since $\mathcal{P}_p^N$ is convex, the preceding inclusion implies

$$\mathcal{P}_q^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_q^N) \subseteq \mathcal{P}_p^N.$$

Equivalently, every finite convex combination of its generators stays in the source body. This is the body inclusion.

3. **Use the inverse isometry for the reverse inclusion.** The bijection $\Delta^{-1} : \Sigma_q \rightarrow \Sigma_p$ satisfies the same chord-distance condition with p, q interchanged. Applying Steps 1–2 to that map gives

$$\mathcal{P}_p^N \subseteq \mathcal{P}_q^N.$$

The two inclusions are for the same output dimension N and the same reference minor coordinates, so they yield the asserted equality.

Main citations

- Exact declaration and its source — `MathlibAnnex.PluckerBody.eq_of_sphereIsometry`
- Generator transport through the sphere isometry — `MathlibAnnex.PluckerBody.normalizedGenerator_mem_of_sphereIsometry`
- Sign symmetry of the source body — `MathlibAnnex.PluckerBody.body_neg`
- the signed-generator inclusion — `MathlibAnnex.PluckerBody.rawGenerators_subset_of_sphereIsometry`
- the body inclusion — `MathlibAnnex.PluckerBody.subset_of_sphereIsometry`
- Signed generator set — `MathlibAnnex.PluckerBody.generators`
- Convex-body definition — `MathlibAnnex.PluckerBody.body`

Lean source signature (exact)

```
/-- Equality follows from the two inclusions furnished by the isometry and its inverse. -/
theorem eq_of_sphereIsometry {m N : ℕ}
  {MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)}
  (Δ : sphere (0 : Space MX) 1 ≈i sphere (0 : Space MY) 1) :
  body MX N = body MY N
```

The two expressions after the final colon are sets, and $=$ asserts that they contain exactly the same minor-coordinate vectors.

In the source	Mathematical meaning
$\{m\ N : \mathbb{N}\}; \text{Fin } (m + 1) \rightarrow \mathbb{R}; \text{Fin } N \rightarrow \mathbb{R}$	$m, N \geq 0, n = m + 1 > 0, E = \mathbb{R}^n, F = \mathbb{R}^N$. Coordinate vectors have the sup norm; braces mark parameters Lean can infer.
$\text{MX}; \text{MY}$	The continuous seminorms $p = \text{MX} \cdot p$ and $q = \text{MY} \cdot p$, with their positive lower and upper comparison constants stated in this Card.
$\text{Space } \text{MX}; \text{Space } \text{MY}$	The same vectors of E , now with norms p and q , respectively.
$\text{sphere } (0 : \text{Space } \text{MX})\ 1; \text{sphere } (0 : \text{Space } \text{MY})\ 1$	$\Sigma_p = \{x : p(x) = 1\}$ and $\Sigma_q = \{x : q(x) = 1\}$. The arguments are center 0 and radius 1.
$\Delta : \dots \approx_i \dots$	A bijective isometry $\Delta : \Sigma_p \rightarrow \Sigma_q$: $q(\Delta u - \Delta v) = p(u - v)$ for all $u, v \in \Sigma_p$. The symbol $\approx_i$ in the code asserts both bijectivity and distance preservation.
$\text{Matrix.MaximalMinorIndex } (m + 1) (\text{Fin } N)$	The row selections $I = (i_1 < \dots < i_n)$ with $n = m + 1$ rows from $\{1, \dots, N\}$. Both bodies lie in the same space $\mathbb{R}^{\mathcal{I}_{n,N}}$.
$\text{Matrix.ballVolumeScaledMaximalMinors } \text{MX } B; \dots$ $\text{MY } B$	$\mathcal{V}_p(B) = v_p \Pi_n(B)$ and $\mathcal{V}_q(B) = v_q \Pi_n(B)$, where $v_s = \lambda_n(\{x : s(x) \leq 1\})$ and $\Pi_n(B)_I = \det((Be_j)_{i_a})_{a,j=1}^n$.
$\text{generators } \text{MX } N$	$\mathcal{G}_p^N = \{\pm v_p \Pi_n(B) : B : E \rightarrow F \text{ linear}, \forall z, \ Bz\ _\infty \leq p(z)\}$, where $v_p = \lambda_n(\{x : p(x) \leq 1\})$.
$\text{generators } \text{MY } N$	$\mathcal{G}_q^N = \{\pm v_q \Pi_n(B) : B : E \rightarrow F \text{ linear}, \forall z, \ Bz\ _\infty \leq q(z)\}$. This uses q and v_q , instead of p and v_p .
$\text{body } \text{MX } N$	The source body $\mathcal{P}_p^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_p^N)$, a set of minor-coordinate vectors.
In the source	Mathematical meaning
$\text{body } \text{MY } N$	The target body $\mathcal{P}_q^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_q^N)$.

In the source	Mathematical meaning
body MX N = body MY N	The conclusion $\mathcal{P}_p^N = \mathcal{P}_q^N$. The same arbitrary output dimension N is used on both sides; no linear extension of Δ is asserted.

Exact definitions used in this reading: Signed generator set; Convex-body definition.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 15–18:

```
noncomputable section
open Set Metric Function MeasureTheory
open scoped NNReal ENNReal BigOperators
namespace MathlibAnnex.PluckerBody
```

▼ Exact content identity

Declaration: MathlibAnnex.PluckerBody.eq_of_sphereIsometry

Accepted content SHA-256: aeded10096a0d780b78a532d3c609f27b4e08e9215fb29d6377b3c4cc99382b5

Accepted source guide SHA-256: aad4287bcfc0e0bfe21426313cdf86c427e7d5db183d4bab61808c77aef4593a

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73